

ALEX YAN

U.S. Citizen | Wexford, PA | 412-699-3595 | alexyan2309@gmail.com | [Linkedin](#) | [Github](#) | [Personal Website](#)

EDUCATION

Purdue University

B.S. Computer Science

West Lafayette, IN

December 2027

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Systems Programming, Computer Architecture, Programming in C, Discrete Mathematics, Linear Algebra, Introduction to Statistics

EXPERIENCE

PNC Bank

Software Engineering Intern

Pittsburgh, PA

Jun. 2026 – Aug. 2026

- Incoming 2026 Summer Intern on the Auto Lending Technology team.

Rooh (Berkeley SkyDeck Pad-13 Incubator)

Software Engineering Intern

Berkeley, CA (Remote)

Jun. 2025 – Dec. 2025

- Engineered a real-time conversational AI app in React and Next.js, orchestrating an end-to-end voice pipeline with Google Cloud STT, LLM dialogue generation, and Google Cloud TTS, reducing response latency by **20%**.
- Architected Next.js backend data pipelines supporting a live AI simulator and async review tool, handling real-time audio streaming, buffering, and transcription at scale.
- Designed a custom React analytics dashboard generating session-based delivery and tone metrics, giving coaches actionable feedback on AI performance.
- Redesigned the marketing landing page with a dynamic scrollytelling component, improving user engagement on the product's primary acquisition surface.

Purdue Stack

Full Stack Developer

West Lafayette, IN

Sept. 2025 – Present

- Shipped a research collaboration platform in Next.js, FastAPI, and PostgreSQL for Purdue's Statistics Department, featuring role-based dashboards, real-time messaging, and JWT authentication. (Spring 2026)
- Delivered a TA assignment optimization system for Purdue's Mathematics Department serving 150–200 TAs per semester, reducing scheduling setup time by **25%** via a preference-weighted greedy matching algorithm in Python. (Fall 2025)

PROJECTS

PyroTech | *Next.js, TypeScript, Google Gemini, WebSocket* | [Github](#)

Apr. 2026

- Developed an AI wildfire incident command simulation platform with a Google Gemini multi-agent orchestration layer coordinating 7 specialized agents across disaster, evacuation, resource, and infrastructure domains.
- Engineered a physics-based fire spread engine running at a **500ms tick rate** with dozer barrier mechanics and suppression zone modeling, streamed live to clients via WebSocket.
- Pulled real-world map data to power layered geospatial visualizations, supporting branching what-if scenario analysis for incident commanders.

Iris | *JavaScript, Chrome MV3, AWS Lambda, OpenAI* | [Github](#)

Feb. 2026

- Developed a voice-driven productivity assistant for Google Workspace at Carnegie Mellon's 24-hour flagship hackathon, targeting routine email, calendar, and document workflows.
- Designed a serverless AWS Lambda backend and Chrome MV3 extension with wake-word detection, real-time audio capture, and LLM-driven intent planning.
- Delivered a fully functional demo supporting natural voice commands across Gmail, Calendar, and Docs within the 24-hour window.

Access Prep | *React, TypeScript, Flask, LangChain, Gemini* | [GitHub](#)

Jan. 2026

- Won **3rd Place** at the Capital One Tech Summit Hackathon by building a bilingual document preparation platform serving underrepresented populations across 5 immigration and financial aid pathways.
- Implemented a Flask REST API handling session management, file uploads, and AI validation, integrating LangChain and Gemini for context-aware, prompt-engineered responses.
- Engineered an OCR document validation pipeline processing 5 document types to verify applicant eligibility automatically.

SKILLS

Languages: Python, GoLang, Java, JavaScript, TypeScript, C, C++, SQL, Bash

Frameworks: React, React Native, Next.js, Flask, Node.js, Express.js, Tailwind CSS

Databases/Cloud: PostgreSQL, MongoDB, AWS (Lambda, Gateway), Google Cloud, Docker

Awards: Eagle Scout, 6x Pennsylvania State Raw Powerlifting Record Setter